LUKE Simulations of LHCD in ITER

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1 Introduction

The LUKE ray-tracing/Fokker-Planck package is used to perform simulations of LHCD in ITER. An up/down symmetric ITER Scenario 4 equilibrium is used. The equilibrium data were provided by P. Bonoli for ITER simulations presented at the IAEA conference in 2006. Figures representing the equilibrium are available in Section 2. In section 3 a scan in $N_\parallel$ is performed for the wave frequencies $f = 5.0$ GHz and $f = 3.7$ GHz, respectively. A poloidal scan of ray lauching positions is presented in section 4. The results are discussed in Section 5.

2 Equilibrium

The magnetic equilibrium is plotted in Fig. 1 and the profiles in Fig. 2.

3 $N_\parallel$ scan at $f = 5.0$ GHz and $f = 3.7$ GHz

For each scan, six rays starting from the outboard side are launched into the plasma and propagated by the ray-tracing code C3PO. The six rays correspond to the positive and negative lobes at the three different vertical launching positions $Z = -0.5/0/0.5$ m. The total power in $P = 30$ MW. The current deposition profiles and total current corresponding to a scan with $N_\parallel = 1.9/2.0/2.1$ are shown in Fig. 3 and 4 corresponding to the frequencies $f = 5.0$ GHz and $f = 3.7$ GHz, respectively. It is found that the current drive efficiency increases for lower $N_\parallel$. This trend is expected since the velocity $v_\parallel = c/N_\parallel$ of resonant electrons is larger for lower $N_\parallel$. We note that the trend holds in all cases except for $f = 3.7$ GHz and $N_\parallel = 1.9$ where the driven current is much lower than expected.

For a given initial value of $N_\parallel$ (except the case $N_\parallel = 1.9$), the current drive efficiency is significantly higher for $f = 3.7$ GHz than for $f = 5.0$ GHz. To interprete this results the rays are showed in Fig. 5 for $N_\parallel = 2.0$ and each frequency. The value of $N_\parallel$ at the location of maximum power deposition is displayed in Fig. 6. We observe that the ray propagation is different for the two frequencies. When $f = 3.7$ GHz, the rays experience more refraction and a larger $N_\parallel$ downshift and thus propagate further into the plasma. This explains the larger current drive efficiency.

In order to explain what happens for $f = 3.7$ GHz and $N_\parallel = 1.9$, the rays are shown in Fig. 7 for $N_\parallel = 1.9$ and each frequency. The value of $N_\parallel$ at the location of maximum power deposition is displayed in Fig. 8. We observe that for $f = 3.7$ GHz two of the three

rays experience so much refraction that they propagate back towards the edge without being damped. After reflection on the edge they experience a very large $N_{\|}$ upshift leading to very inefficient current drive localized close to the edge.

4 Poloidal scan

Rays were launched from 16 different, uniformly distributed, poloidal locations with $f = 5.0$ GHz and initial $N_{\|} = 2.0$. These rays are shown on Fig. 9. Rays are counted counted-clockwise starting with the ray launched from the horizontal outboard midplane. On Fig. 10 and Fig. 11, the damping radial location and corresponding $N_{\|}$ value are displayed, respectively, as a function of the ray number. It is found that rays lauched from the horizontal midplane or slightly above (either on the LFS or the HFS) experience a larger downshift and propagate further into the plasma than rays lauched from the top or bottom of the plasma. Because these rays at absorbed at a lower value of $N_{\|}$ they also have the potential for more efficient current drive.

5 Discussions and Conclusions

Concerning the poloidal location of ray launching, it is demonstrated that rays experience a larger $N_{\|}$ downshift and thus propagate further into the plasma if they are launched on or slightly above the horizontal midplane. There are no significant differences between LFS and HFS launching.

For a given frequency, rays launched with a lower $N_{\|}$ propagate further into the plasma and lead to a larger current drive efficiency. This trend holds when lowering $N_{\|}$ until the rays are refracted back to the edge before being damped, in which case the contribution of such ray to driving current drops considerably. For a given initial $N_{\|}$, rays launched at the frequency $f = 3.7$ GHz experience a stronger downshift and thus drive more current than with $f = 5.0$ GHz. This also means that the limit in $N_{\|}$ for which the ray is reflected back to the edge before being damped, which must be avoided, is higher in the case where $f = 3.7$ GHz.

More generally, this limit in $N_{\|}$ results from the balance between refraction - which depends upon the density profile - and damping - which depends upon the temperature profile. The objective is to keep $N_{\|}$ high enough such that the ray is damped before being refracted, but low enough to maintain a good current drive efficiency.

The maximum current drive efficiency that can be obtained is similar with $f = 3.7$ GHz and with $f = 5.0$ GHz, but it occurs at a higher value of $N_{\|}$ in the $f = 3.7$ GHz case.

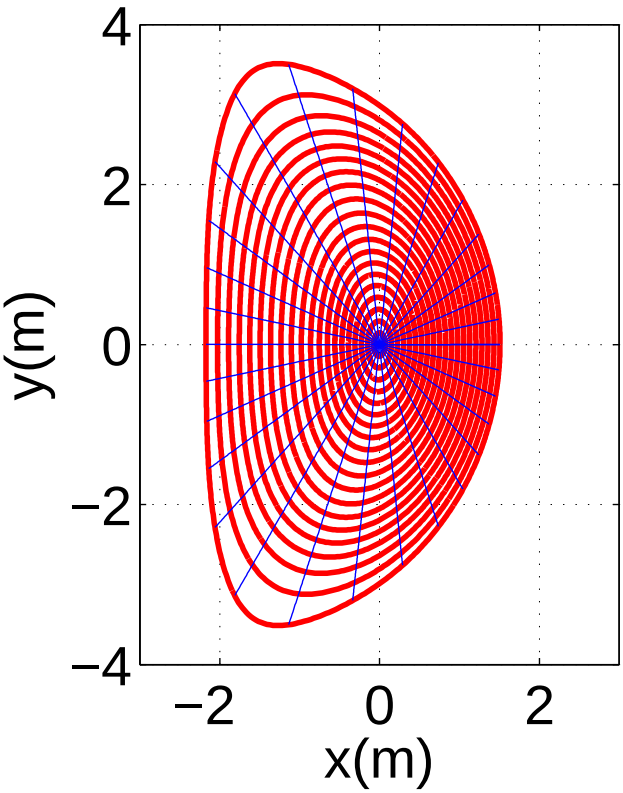


Figure 1: Magnetic equilibrium of ITER scenario 4. The red and blue lines are contours of constant ψ and θ, respectively.

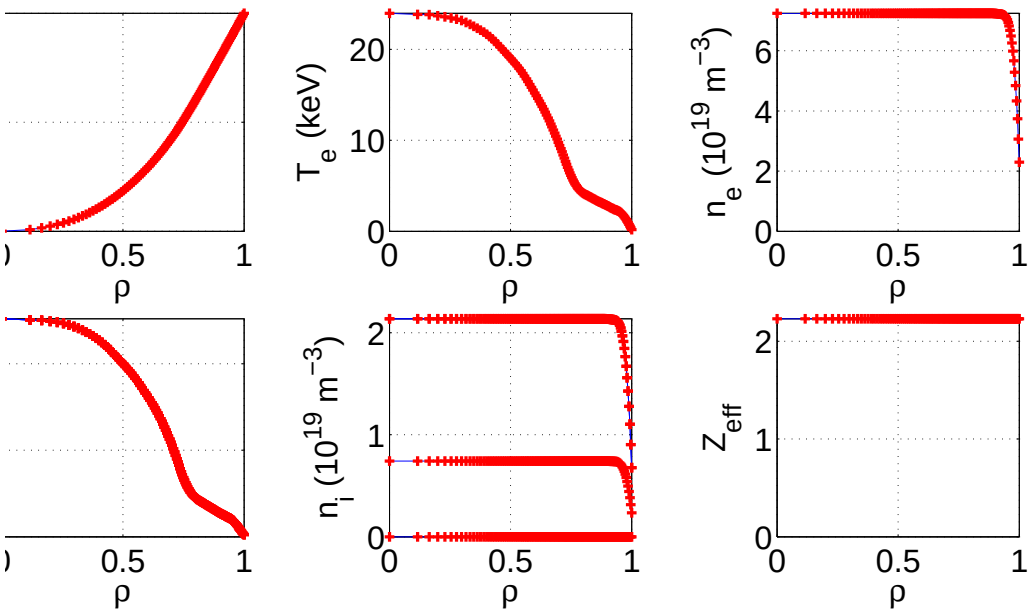


Figure 2: Radial profiles of normalized poloidal magnetic flux, electron temperature, electron density, ion temperatures, ion densities, and effective charge for the ITER Scenario 4 equilibrium.

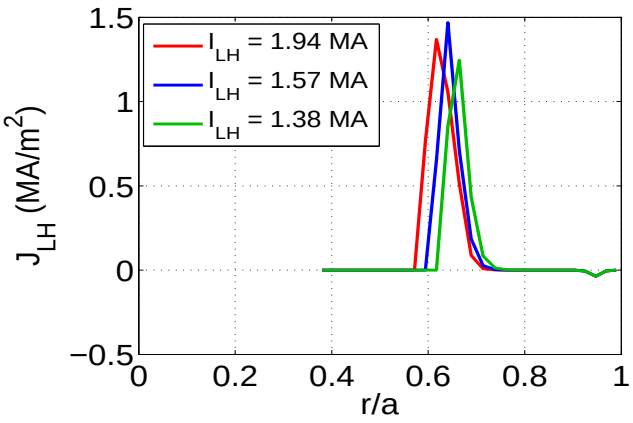


Figure 3: Current density profile and total driven current for $f = 5.0$ GHz and $N_{\|} = 1.9$ (red), $N_{\|} = 2.0$ (blue), and $N_{\|} = 2.1$ (green).

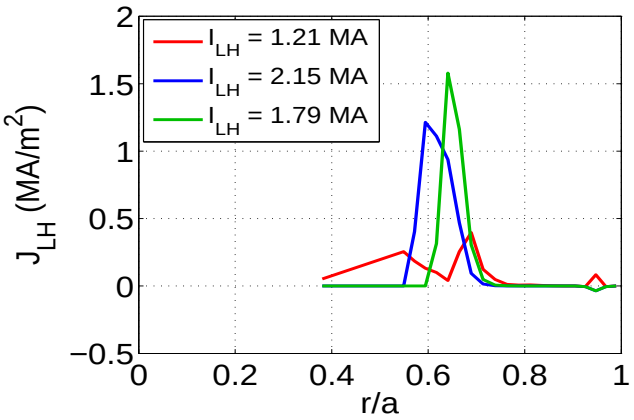


Figure 4: Current density profile and total driven current for $f = 3.7$ GHz and $N_{\|} = 1.9$ (red), $N_{\|} = 2.0$ (blue), and $N_{\|} = 2.1$ (green).

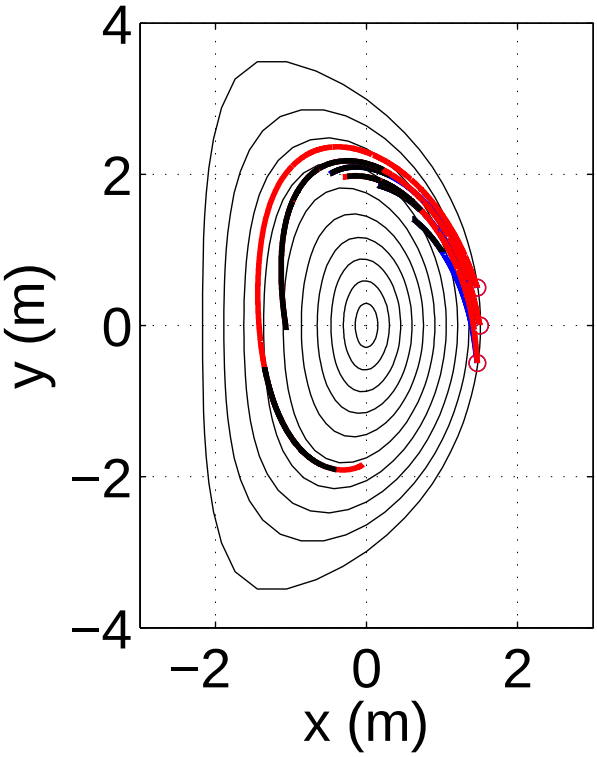


Figure 5: Poloidal ray trajectories of the three rays corresponding to the positive lobe and three different poloidal launching positions. The rays are calculated for $N_{\|} = 2.0$, and $f = 5.0$ GHz (blue) or $f = 3.7$ GHz (red).

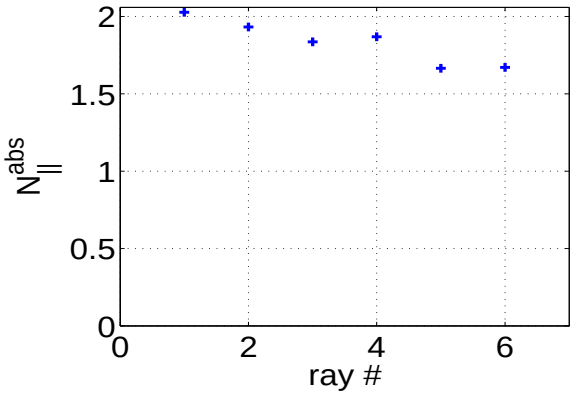


Figure 6: Value of $N_{||}$ at the location of maximum power deposition for the six rays shown in Fig. 5. Rays #1,2,3 correspond to $f = 5.0$ GHz and $Z = -0.5/0/0.5$ m, while rays #4,5,6 correspond to the same vertical launching positions with $f = 3.7$ GHz.

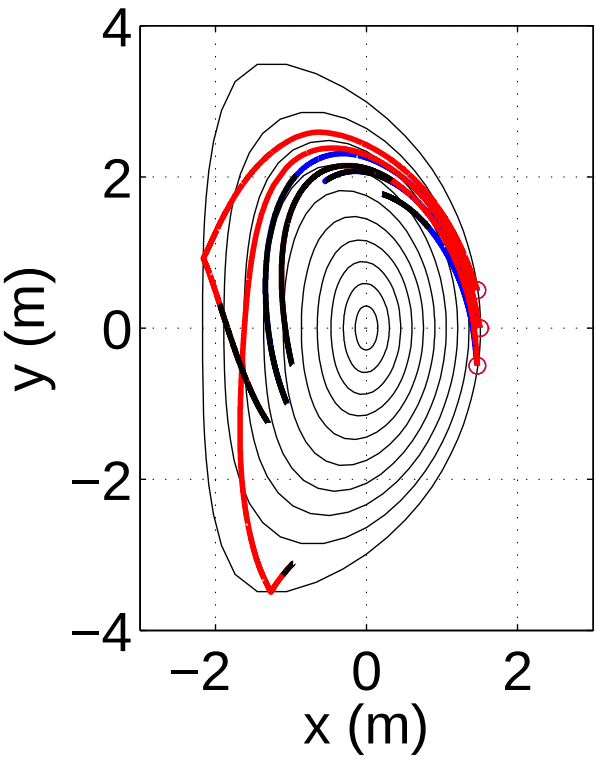


Figure 7: Poloidal ray trajectories of the three rays corresponding to the positive lobe and three different poloidal launching positions. The rays are calculated for $N_{||} = 1.9$, and $f = 5.0$ GHz (blue) or $f = 3.7$ GHz (red).

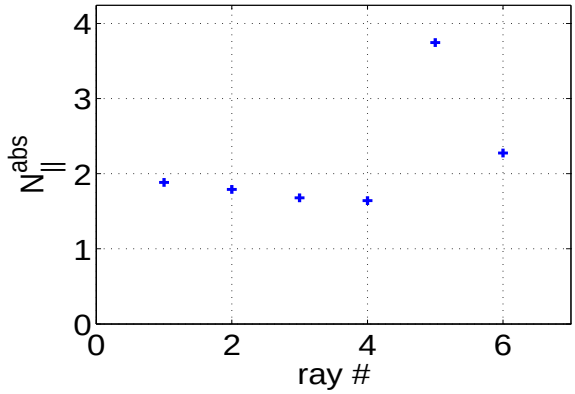


Figure 8: Value of $N_{\parallel}$ at the location of maximum power deposition for the six rays shown in Fig. 7. Rays #1,2,3 correspond to $f = 5.0$ GHz and $Z = -0.5/0/0.5$ m, while rays #4,5,6 correspond to the same vertical launching positions with $f = 3.7$ GHz.

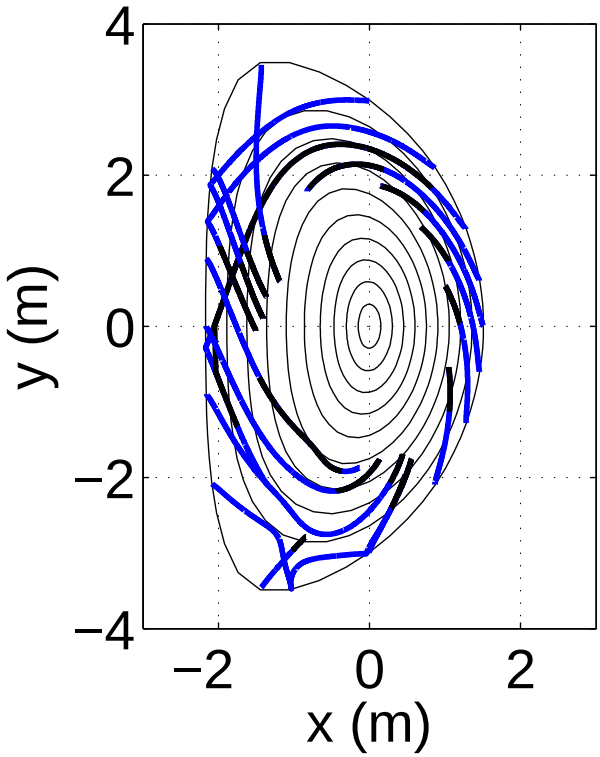


Figure 9: Poloidal trajectories of LH rays. The black lines represent the ray portion where damping is significant.

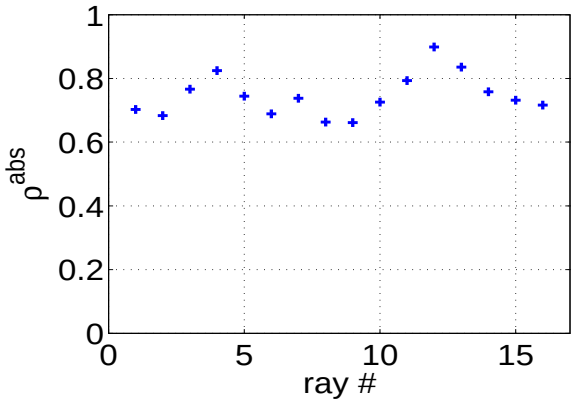


Figure 10: Radial position of the peak location ρ^{abs} in the power deposition profile. $\rho^{\mathrm{abs}}(\psi)$ corresponds to r/a where the flux surface ψ crosses the outboard horizontal midplace.

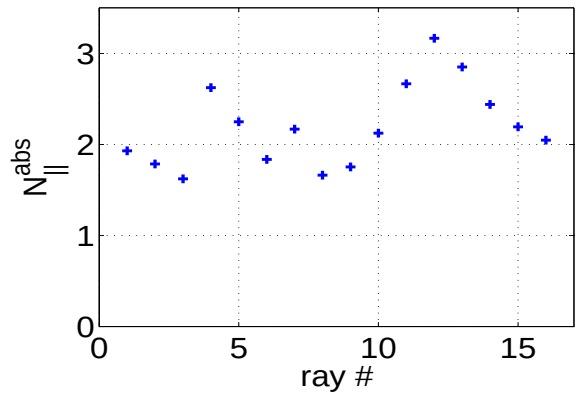


Figure 11: Value of $N_{\|}$ along the ray at the position of maximum damping ρ^{abs}.